

Saideep Tiku

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RELEVANT ROLES & EXPERIENCE

• Micron Technology

- **Principal Systems Architect** (Dec 2025, Ongoing)
- **Senior Systems Architect** (May 2023 – Nov 2025, ~2.5 years)
- **Systems Architect** (Mar 2022 – May 2023, ~1 year)
 - Influence strategy for next-gen memory concepts with relevant stakeholders up to c-suite
 - Drive AI workload analysis with cross domain teams to generate technical memory product requirements
 - Engage in strategic partnership with lighthouse embedded customers and product-specific chipset vendors
 - Generate technical collateral on cutting edge technologies such as XR, ADAS, and Humanoid Robotics with a special focus on their impact on memory and storage systems and technologies
 - Impacting technical specification of next-generation products and market segment strategies
 - Serve as a consultant to Micron's venture investment team on emerging opportunities/activities
 - Filed patents on efficient deployment of AI workloads on various compute and memory sub-systems
 - Develop/Propose tools for AI performance and power estimation on next-gen memory products
 - Interface internal teams specializing in systems simulations, product arch., field apps, system engineering
 - Author customer facing tools to market memory for AI (**published at internal technical conference**)
 - Deliver technical presentations at industrial conferences such as FMS, Electronica, Embedded Vision, etc.
- **Machine Learning Architect Intern** (January 2020 – January 2021, 1 year)
 - System level feasibility analysis of emerging memory technology for smartphones
 - Filed patents on aging resilience, and improving energy-efficiency of AI workloads for compute and memory
 - Evaluate and analyze DNNs in terms of performance, memory requirements, power consumption, area, complexity, and flexibility on hardware platforms such as Google's Edge TPU, Intel's NCS2, NVIDIA's DLAs
 - Technical venture analysis for Micron's capital investment opportunities/activities among startups
 - Coauthored work on edge AI accelerators at "Systems Solutions Technical Conference" (**internal conference**)
- **Mentor Graphics: Calibre DesignRev Enhancement (Recon mode)** (June 2019 – July 2019, 8 weeks)
 - Designed a rapid debug methodology that enables intelligent physical verification at design time
 - Developed a machine learning-based ASCII Results Database (RDB) debug and visualization interface
- **Fiat Chrysler Automobiles and Colorado State University (Joint)** (Aug. 2015 – Dec. 2016, 1 year, 6 months)
 - Exploration and automation of data/code towards the migration to a multi-core powertrain platform
- **Cryptotronix: Embedded Systems Intern** (Available on GitHub) (Jan. 2015 - Apr. 2015, 4 Months)
 - Maintained kernel module and tested drivers for Atmel crypto chips (ATSHA204 and ATECC108)

ACADEMICS

- **Ph.D.** in Electrical & Computer Engineering at Colorado State University
 - **Research Areas:** Machine Learning, AI, Mobile Computing, Embedded Systems, Computer Architecture

TECHNICAL SKILLS

- Platforms: Python (Matplotlib, Pandas, Numpy), C, C++, SystemC, OpenMP, CUDA, R, and Java (Android)
- Skillset: Machine Learning (Sklearn, Caffe2, Tensorflow), Predictive Analysis, Deep Learning
- Cross-domain: Gem5, McPat (Performance Analysis), Quantization and Compression (TF Lite)

KEY PUBLICATIONS AND PATENTS / ACADEMIC ACHIEVEMENTS

- S. Tiku, P. Kale, "Artificial Neural Network Training in Memory", US Patent No. 11556790, 2023
- S. Tiku, S. Pasricha, "Siamese Neural Encoders for Long-Term Indoor Localization with Mobile Devices", to appear, ACM/IEEE Design, Automation and Test in Europe Conference (DATE), 2022
 - Proposed a Triplet-loss centric Siamese encoder for aging resilient indoor localization
- S. Tiku, P. Kale, S. Pasricha, "QuickLoc: Adaptive Deep-Learning for Fast Indoor Localization with Mobile Devices", ACM Transactions on Cyber-Physical Systems (TCPS), 2021
 - Developed deep-learning models employing early-exit for efficient indoor localization
- S. Tiku, S. Pasricha, "SARTHI: A Secure, Accurate, Real-Time, and Heterogeneity-Resilient Indoor Localization Framework for Smartphones," (PhD Forum) Design Automation Conference (DAC), 2019 (**Best Poster Award**)
 - Presented an overview of my ongoing work in the area of indoor localization at DAC's PhD forum event
- S. Tiku, S. Pasricha, "Overcoming Security Vulnerabilities in Deep Learning Based Indoor Localization on Mobile Devices," to appear, ACM Transactions on Embedded Computing Systems (TECS), 2019 (**Patent Filed**)
 - Designed a generalized approach to deploy CNN models resilient to malicious snooping and jamming attacks
- A. Mittal, S. Tiku, S. Pasricha, "Adapting Convolutional Neural Networks for Indoor Localization with Smart Mobile Devices," ACM Great Lakes Symposium on VLSI (GLSVLSI), 2018 (**Paper Award**)
 - Proposed a novel CNN-based approach for indoor localization using smartphone Wi-Fi